

AYE
TECH HUB

• AI & PRODUCTIVITY • LESSON 07

NotebookLM for Research

Your AI Research Assistant Trained on Your Sources

Lesson 07 of 30 | AI & Productivity Program

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- How NotebookLM differs from ChatGPT — why it never hallucinates your sources
- Adding sources: PDFs, Google Docs, YouTube links, websites and audio
- The inline citation system — every answer traced to a specific source
- Audio Overview: turn 200 pages into a 20-minute podcast in one click
- 10 professional research workflows using NotebookLM for engineers

01 | What Is NotebookLM — and Why Is It Different?

NotebookLM (notebooklm.google.com) is a free AI research assistant from Google. Unlike ChatGPT, Claude, or Gemini — which answer from their vast training data and can hallucinate — NotebookLM works **exclusively from the sources you upload**. It reads your documents, and only your documents. Every answer includes an inline citation to the exact source and paragraph it drew from.

ChatGPT / General LLMs	NotebookLM
Trained on internet data (up to training cutoff)	Trained ONLY on sources you upload to each notebook
Answers from training data (may hallucinate facts)	Answers ONLY from your sources (no hallucination)
Cannot cite a specific page or paragraph	Every answer has an inline citation to your source
May confuse your document with training data	Cannot answer beyond your uploaded sources

Figure 1 — NotebookLM vs General LLMs: The Key Difference

Feature	NotebookLM	ChatGPT / Claude / Gemini
Training data source	ONLY your uploaded documents	Vast internet dataset (training cutoff)
Hallucination risk	Near zero — only quotes from sources	Present — may invent plausible-sounding facts
Citation system	Inline citation: every answer cites source + page/paragraph	None (most) or incomplete citations
Maximum sources	50 sources per notebook	Limited by context window (~20-200 pages depending on model)
Audio output	Yes — generates podcast-style Audio Overview	No native audio generation
Best for	Deep research on specific documents, technical standards, reports	General questions, creative tasks, broad knowledge

The No-Hallucination Guarantee

When you ask NotebookLM "What are the load combination requirements?" and it answers, it **ONLY** uses information from the standards you uploaded. If the answer is not in your sources, it tells you: "I cannot find this information in the provided sources." This is the exact opposite of ChatGPT's behaviour — and for professional engineering work, it is invaluable.

02 | Setting Up Your First NotebookLM Notebook

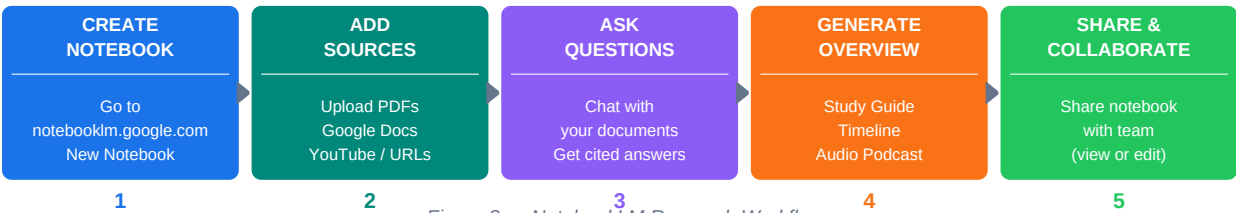


Figure 2 — NotebookLM Research Workflow

Step-by-Step: Your First Engineering Notebook

Step	Action	Notes
1 — Go to NotebookLM	Open notebooklm.google.com in Chrome. Sign in with your Google account.	Free plan: 50 sources per notebook, 100 notebooks total.
2 — Create a new notebook	Click "New Notebook" → give it a name (e.g. "Solar PV Design Project")	Name notebooks by project or topic — not by date.
3 — Add your first sources	Click "+ Add source" → choose: Upload (PDF), Google Drive, website URL, YouTube link, or paste text.	NotebookLM processes each source and indexes it within 30-60 seconds.
4 — Start the chat	Type your first question in the chat box on the right. "What are the main requirements for solar panel mounting according to these documents?"	The answer appears with inline citations — click any citation to jump to the source.
5 — Generate notebook guide	In the "Notebook guide" panel → click "Generate". NotebookLM creates: Summary, Key Topics, Study Questions, Table of Contents, Timeline (if applicable)	This takes 30-60 seconds and gives a complete overview of all your sources.

6— Ge ner ate Aud io O ver vie w	Notebook Guide panel → click "Generate" under Audio Overview. Two AI hosts discuss your documents in a podcast-style conversation.	Takes 2-5 minutes. Can be downloaded as MP3. Excellent for commuting or learning on the go.
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Source Types NotebookLM Accepts

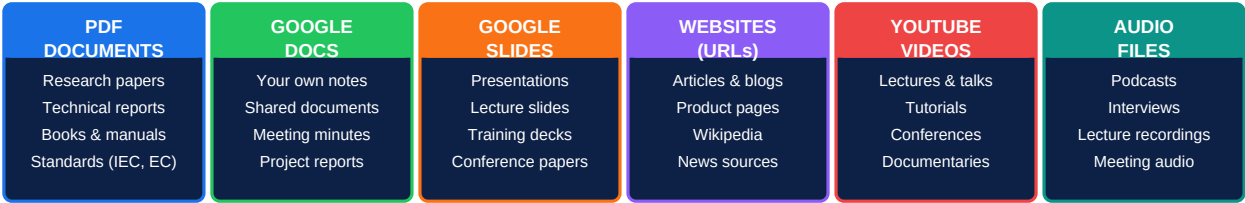


Figure 3 — Source Types NotebookLM Can Analyse

Source Type	Max Size	Best Practice
PDF document	500MB per file	Use text-based PDFs (not scanned images) for best results
Google Docs/Slides	No size limit (live sync)	Changes sync automatically — great for ongoing notes
Website URL	Single page only	For multi-page sites, add each important page as separate source
YouTube video	Any length	NotebookLM reads the transcript — quality depends on caption accuracy
Copy/paste text	Up to 500,000 characters	Useful for email threads, chat logs, short articles
Audio (MP3/M4A)	Up to 3 hours	Transcribed automatically — great for lecture recordings

The Citation System — Your Proof Against Mistakes

Every NotebookLM answer shows small numbered citations [1], [2], [3]. Click any number and the source panel jumps to the exact paragraph that was quoted. This means you can verify every statement before including it in a report. This is not just convenient — for professional engineering work where you sign off on documents, it is your audit trail.

03 | Advanced Features and Power Techniques

Feature	How to Use	Best Application
FAQ Generator	Notebook Guide → FAQ → Generate. Creates 10-15 Q&A; pairs from your sources.	Client handouts; staff training; understanding complex standards
Study Guide	Notebook Guide → Study Guide → Generate. Creates learning objectives and key concepts.	Preparing for a professional exam; onboarding new team members
Timeline	Notebook Guide → Timeline → Generate. Extracts chronological events from sources.	Project history; regulatory development timeline; incident analysis
Briefing Document	Notebook Guide → Briefing Doc → Generate. Creates executive summary of all sources.	Board presentations; client briefings; project status reports
Multi-source synthesis	Add documents from different sources and ask comparative questions: "How does EC2 compare to ACI 318 for load combination requirements?"	Standard comparison; cross-referencing multiple project documents
Inline note-taking	Click the pencil icon next to any citation to add your own annotation. Your notes are saved with the source reference.	Building a personal knowledge base with linked evidence
Sharing notebooks	Share icon → choose view or edit access → share link. Collaborators see all sources and can add their own questions.	Team research; client review of technical analysis

04 | 10 Engineering Research Workflows

#	Workflow	Sources to Add	Questions to Ask
1	Understand a technical standard	IEC/ISO/ASHRAE standard PDF	"What are all requirements for [specific topic]? List with clause references."
2	Compare two standards	EC2 + ACI 318 (or any two)	"Compare load combination requirements. What are the key differences?"
3	Analyse a project specification	Client specification, tender doc	"List all mandatory requirements. What are the contractor's obligations?"
4	Research a technical topic	3-5 research papers or textbook chapters	"Summarise the key findings. What is the current consensus on [topic]?"
5	Prepare a client briefing	Project reports, meeting minutes	"Create a briefing document summarising the project status and key decisions."
6	Study for a professional exam	Syllabus + textbook chapters	"Generate 20 exam-style questions with answers. Focus on [topic]."
7	Onboard a new team member	Project specification, drawings, meeting minutes	"Summarise the project background, key decisions, and current status."
8	Create training material	Technical manuals, standards	"Create a training FAQ for a junior engineer on [topic]."
9	Tender/bid research	Previous similar project reports, cost databases	"What were the key cost drivers and lessons learned on similar projects?"

10	Audio learning on the go	Any 5+ relevant PDFs	"Generate Audio Overview" → download and listen while commuting
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What comes next — AI-008

AI Image Generation — how Midjourney, DALL-E, Adobe Firefly, and Stable Diffusion work, how to write effective image prompts, legal and ethical use of AI images, and practical applications for engineers and designers.